

Bagging :-

Bagging, short for bootstrap aggregating, is a machine learning ensemble method designed to improve the stability and accuracy of machine learning algorithms used in statistical classification and regression. It also helps to avoid overfitting. The key principle of bagging is to generate multiple subsets of the original data (with replacement), train a separate model for each subset, and then combine the results.

Defⁿ:-

Random Forest is a versatile and widely used machine learning algorithm that belongs to the class of ensemble methods. Specifically, it is a type of bagging technique, which involves training many individual models (in this case, decision trees) and combining their outputs to make a final prediction.

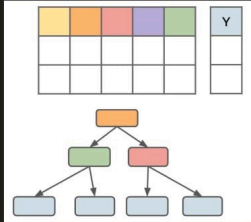
Q2: Explain the "Random" in Random Forest. What are the two main sources of randomness?

A: The "Random" is crucial and comes from two key techniques that de-correlate the trees:

1. **Bootstrap Aggregating (Bagging):** Each tree in the forest is trained on a different random subset of the original training data. This subset is drawn **with replacement**, meaning some data points may be repeated, and others may be left out (these are called "Out-of-Bag" or OOB samples).
2. **Feature Randomness (Random Subspace Method):** When splitting a node in a decision tree, the algorithm is not allowed to search through all features for the best split. Instead, it must choose from a random subset of features (e.g., $\sqrt{n_{\text{features}}}$ or $\log_2(n_{\text{features}})$). This forces the trees to be different and reduces the chance that one very strong feature will dominate all the splits.

Q. Why not continue to use Decision Trees?

- Decision tree restricted by gini impurity.



- No guarantee of using all features.
- Root Node will always be same.
- Root feature has huge influence over tree.

Q3: What is the high-level training algorithm for a Random Forest?

A:

For $b = 1$ to B (where B is the number of trees):

1. Draw a **bootstrap sample** (random sample with replacement) of size N from the training data.
2. Grow a decision tree on this bootstrap sample. At each node:
 - a. Select m features at random from all p features.
 - b. Pick the best split among these m features (e.g., using Gini Impurity or Information Gain).
 - c. Split the node into two daughter nodes.
3. Repeat until the tree is fully grown (no pruning typically).

Output the ensemble of trees: $\{T_b\}_{b=1}^B$

For prediction:

- **Classification:** Majority vote from all trees.
- **Regression:** Average of the predictions from all trees.

Q5: How does a Random Forest handle overfitting?

A single decision tree, if grown deep enough, is highly prone to overfitting; it learns the training data too well, including the noise. Random Forest mitigates this in two primary ways:

1. **Averaging Predictions:** By building many trees and averaging their predictions, the errors of individual trees tend to cancel each other out. The "wisdom of the crowd" helps to smooth out the predictions and reduce the overall variance of the model.
2. **Decorrelating Trees:** The random selection of features at each split ensures that the individual trees are not highly correlated. If all trees were similar, they would make the same errors, and the ensemble wouldn't be as effective. By forcing each tree to consider a different subset of features, the model becomes more diverse and less likely to overfit to specific features in the training data.

Q4: Why is Random Forest better than a single Decision Tree?

A: Random Forests correct for the main weakness of Decision Trees: **overfitting**.

- **Single Decision Trees:** They have high variance. A small change in the data can result in a completely different tree structure. They tend to learn the training data "too well," including its noise, and perform poorly on unseen data.
- **Random Forests:** By building many trees on bootstrapped data and averaging them, they **reduce variance** significantly without increasing bias too much. This leads to a model that generalizes much better to new data.

Q5: Explain the Bias-Variance Trade-off in the context of Random Forests.

A:

- **Bias:** The error due to overly simplistic assumptions in the model. A single decision tree (grown deep) typically has **low bias**.
- **Variance:** The error due to the model's sensitivity to small fluctuations in the training data. A single decision tree has **high variance**.

How Random Forest helps:

It is a **variance-reduction** technique via bagging and feature randomness.

- **Bagging:** By averaging multiple models, the variance of the ensemble is lower than the variance of its individual components. The overall bias remains roughly the same as a single tree.
- **Result:** The forest maintains the **low bias** of deep trees while drastically **reducing the high variance**, leading to a better overall model.

Out-of-Bag samples :-

- **Definition:** Due to bootstrapping (sampling with replacement), each tree is trained on roughly $(\frac{2}{3})$ ~63.2% of the original data. The remaining ~36.8% that are *not* used for training a specific tree are called its **Out-of-Bag (OOB) samples**.

out-of-Bag samples can be used as a built-in validation set.

- **OOB Score:** For each data point, we can gather predictions only from the trees that did *not* have it in their bootstrap sample. The aggregate of these predictions (majority vote or average) gives an **OOB prediction** for that point. The accuracy/ R^2 of all OOB predictions is the **OOB Score**.
- This provides a nearly unbiased estimate of the test set accuracy without the need for a separate validation set, making model evaluation very efficient.

The average errors of these OOB predictions across all data points is called OOB error.



